

Lesson 5: Support Vector Machines



5.0 Lesson Overview



Lesson Overview

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5.1 Building a Default Support Vector Machine Model



Support Vector Machines as Classifier Models



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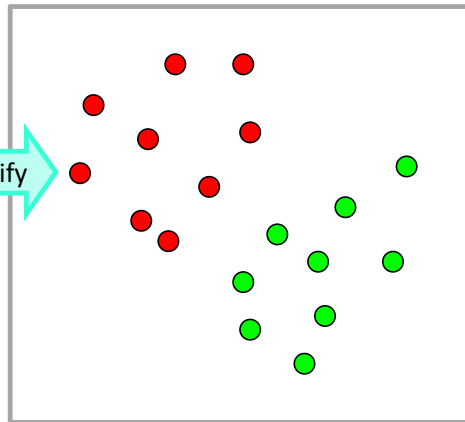
Support Vector Machines as Classifier Models

Support Vector
Machine



Classify

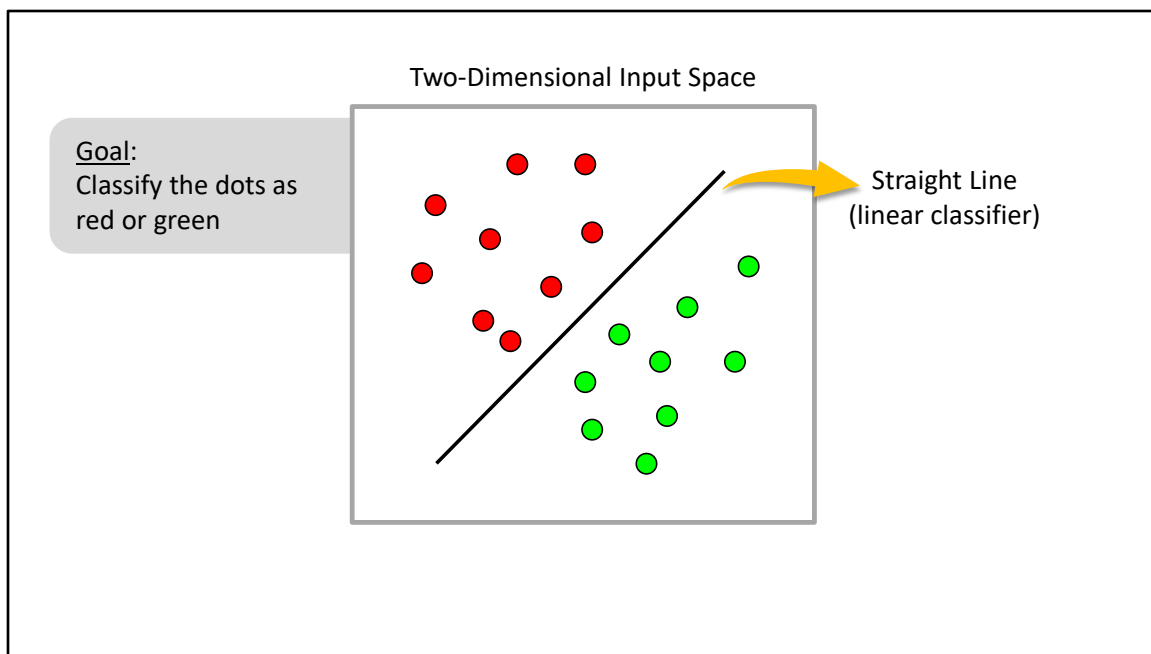
- decision
- rank
- estimate



Like decision trees, support vector machines were originally developed simply to classify outcomes. In other words, the model makes decision predictions instead of ranks or estimates.

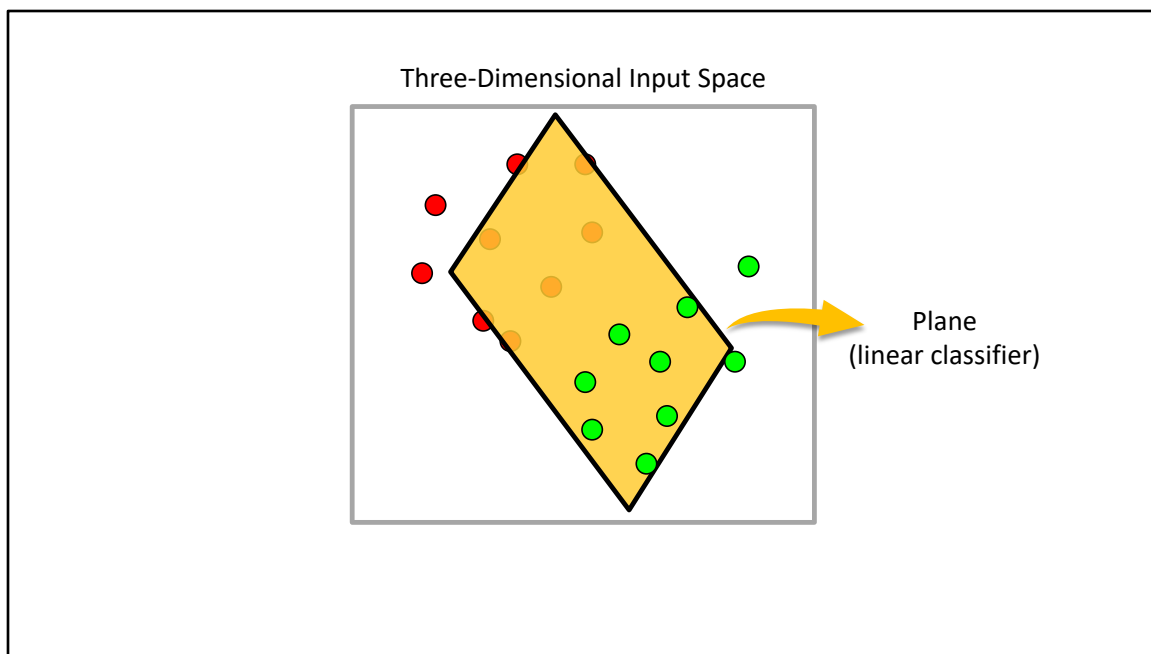
In the basic example shown here, a support vector machine separates the outcomes of a binary target into two classes, red and green.

More recently, support vector machine models have expanded in many ways. Now they can be used for regression tasks to produce estimates. For classification, support vector machines can now provide ranks and estimates. The SVM node in Model Studio fits a support vector machine via interior-point, active-set, or coordinate descent optimization for a binary or interval target. Binary target is the primary focus in this course.

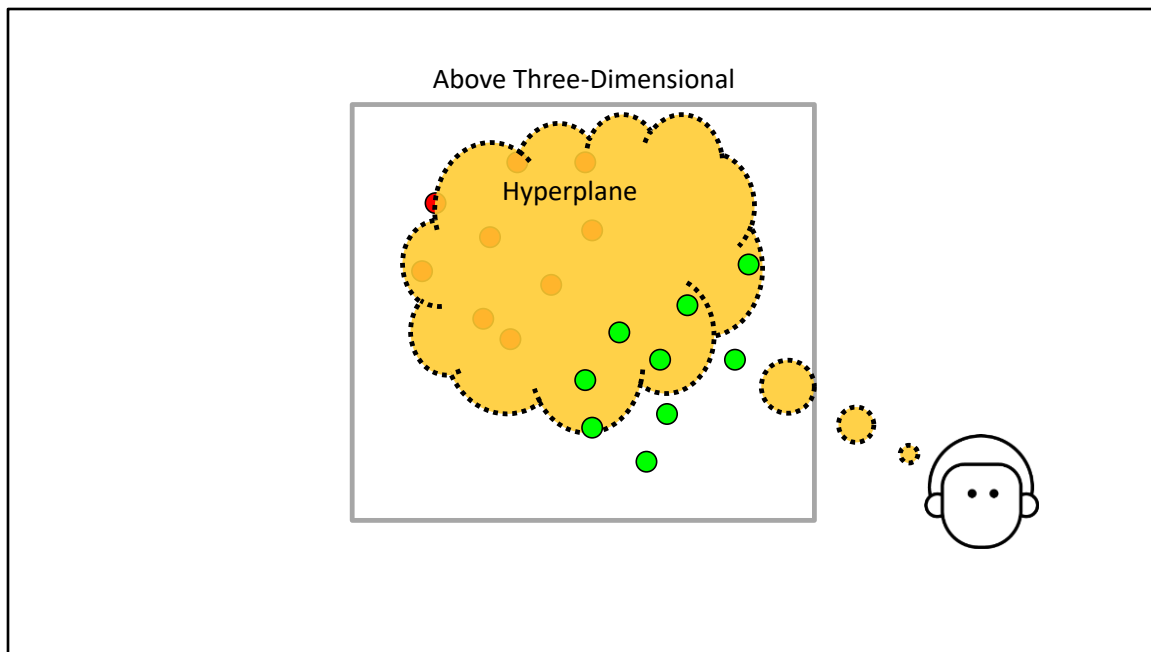


To learn how support vector machines work, we'll focus on this simple, red-versus-green decision case. In this example, the training data set has two inputs, so this is a two-dimensional input space. The goal is to classify the dots as red (that is, the dots in the upper left area) or green (that is, the dots in the lower right area).

The binary outcome is linearly separable, which means that it is possible to draw a line that perfectly separates the target outcomes into two classes (that is, with no errors). This line is the support vector machine. It can also be called a classifier, a classifier model, or a classification rule.

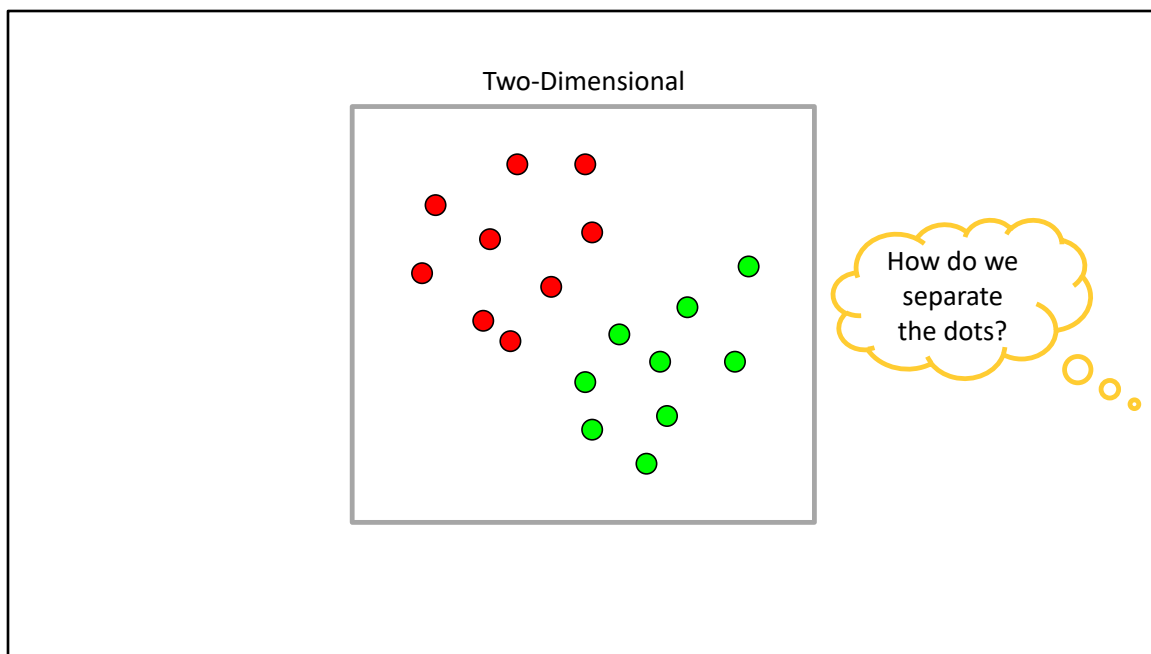


In a three-dimensional input space (that is, when there are three input variables), the classifier is a plane instead of a line. Notice that this diagram shows how the classifier changes but, for simplicity, the third input variable is not shown.



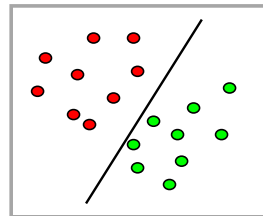
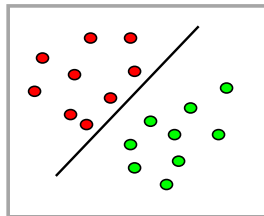
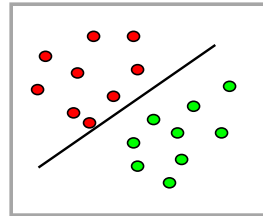
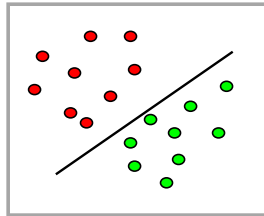
With more than three variables, the classifier is a hyperplane. The diagram represents this concept, but the classifier and additional inputs are not shown.

The techniques covered in this lesson apply to problems of two or more dimensions, so keep in mind that the term hyperplane is sometimes used generally.



Returning to our two-dimensional example, how do we separate the red and green dots?

infinitely
many lines
that perfectly
separate the
target
outcomes ...



Is there
a best
line?

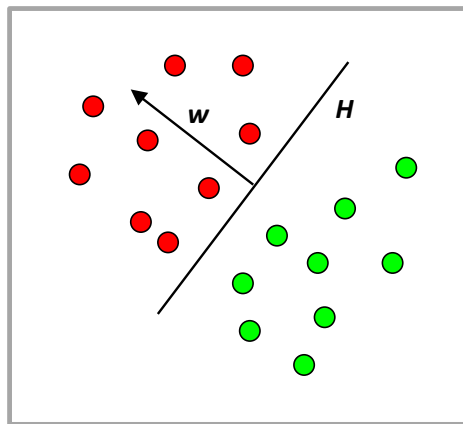
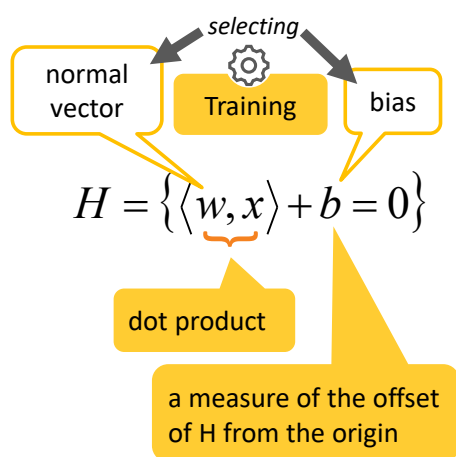
Clearly a line would work. But which line? If the dots are linearly separable, there are infinitely many lines that separate the outcome with no errors. Of the infinitely many solutions, is there a best line?

Mathematical Definition of a Support Vector Machine



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Mathematical Definition of a Support Vector Machine

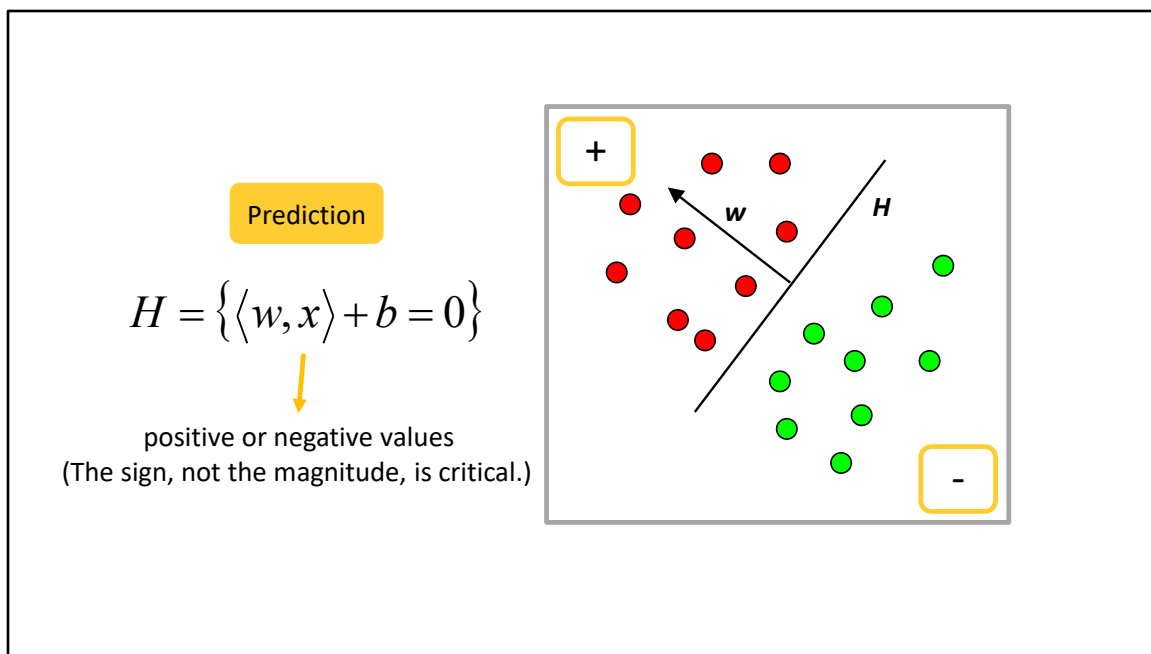


Finding the best solution to a linear classification problem is a mathematical problem, so let's look at the mathematical definition of a support vector machine model. The classifier model (that is, the separating line) is symbolically denoted uppercase H . H has two elements: a normal vector w and a bias term b . The vector w is perpendicular to H . In a two-dimensional input space, where H is a line, w affects the slope of the line. Imagine w as a handle that rotates and affects the flatness or steepness of the line.

The bias term b is a measure of the offset of the separating line from the origin.

H is defined as the dot product between the vector w and the vector of inputs x , plus b . When a point falls exactly on the line, the value of H is 0. A dot product is one way to multiply vectors. A dot product returns a scalar, or single number, as the answer.

To train a support vector machine model, we select w and b so that the line separates the values of the target. (This also applies to a higher-dimension example, which has a hyperplane instead of a line.)



After training, how does the model make predictions for new observations? The formula for H returns predictions as positive or negative values, where the sign, not the magnitude, is the critical aspect. Points on one side of the line (the points in the direction of the normal vector) return a positive value. Points on the other side of the line return a negative value.

Dot Products

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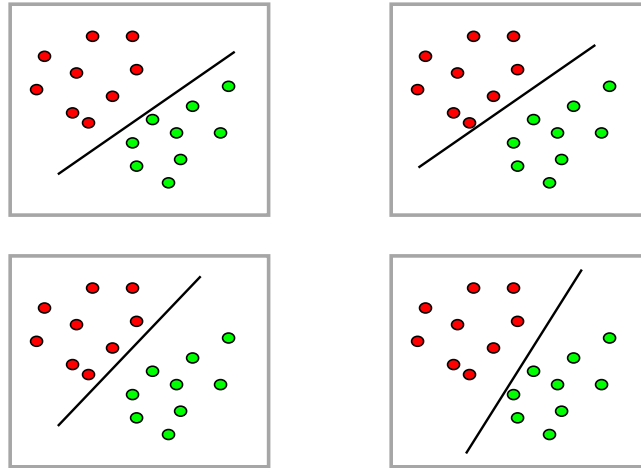


Maximum-Margin Hyperplane and Support Vectors

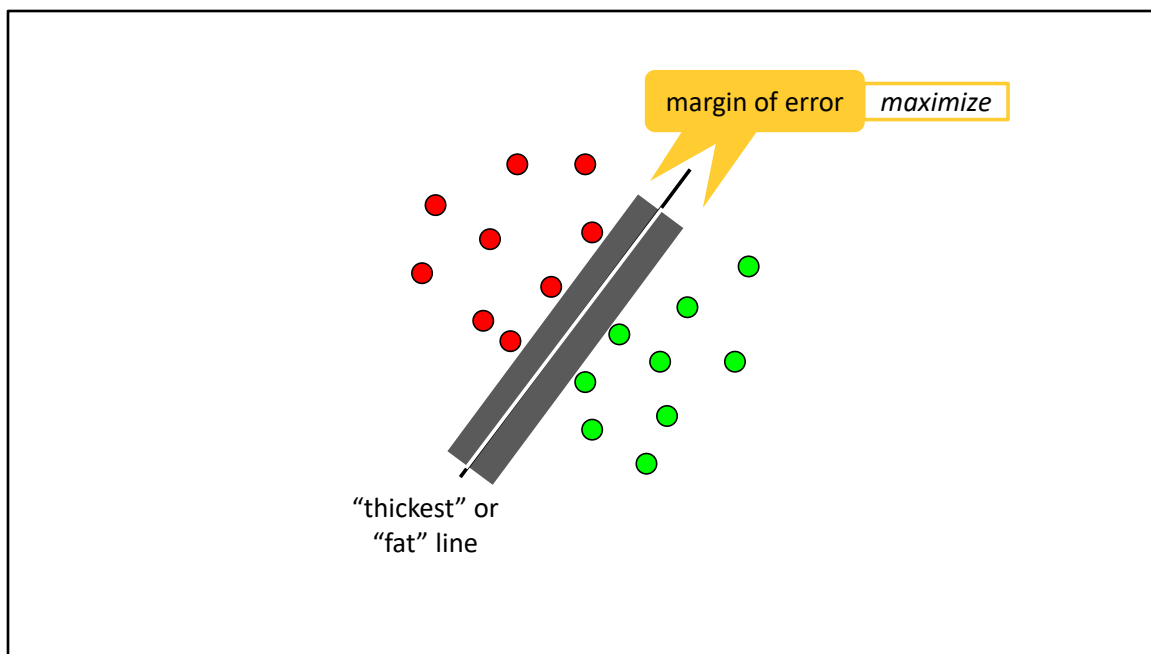


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Maximum-Margin Hyperplane and Support Vectors



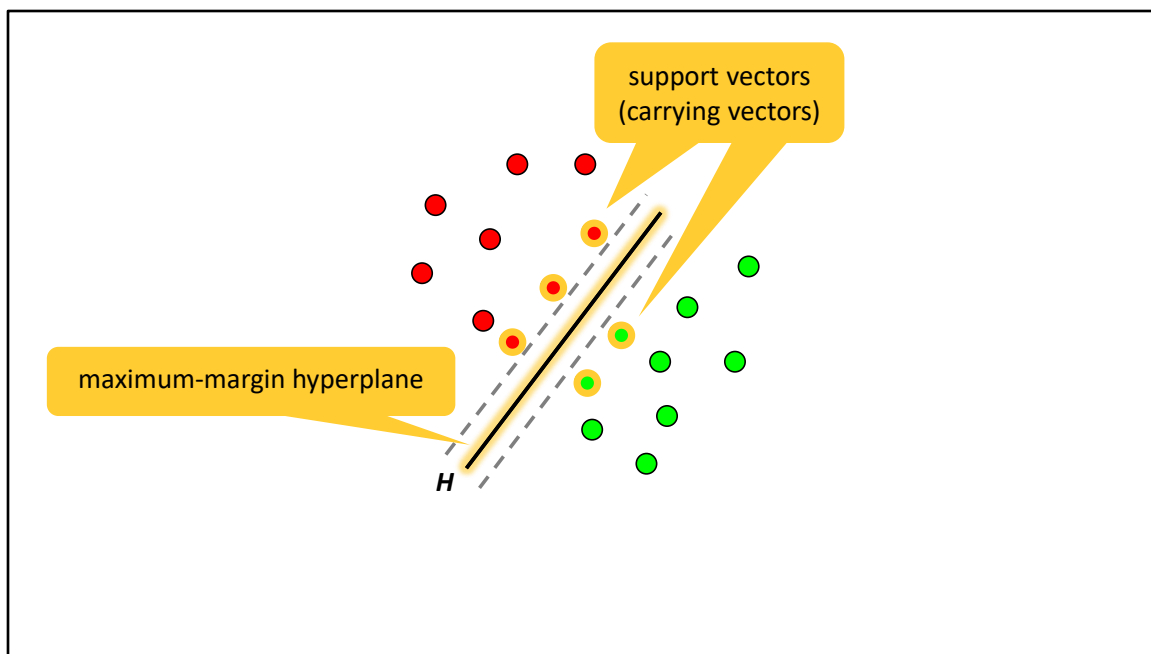
To identify the best classifier for linearly separable data, a good approach is to find the maximum-margin hyperplane.



Let's see how this works. We want to maximize the margin of error on either side of the separator.

In our two-dimensional example, visualize the thickest line that touches the innermost red values of the target and the innermost green values of the target. The thickest (or fat) line is a good starting place for the "best" line because it results in the largest margin of error on the positive and negative sides.

The best solution is the exact center, or median, of the fat line.



Taking the exact center of the fat line produces a unique solution known as the maximum-margin hyperplane. This solution, which is represented as H in the diagram, has the greatest margin of error on either side.

Support vectors, also known as the carrying vectors, are the points in the data that are closest to the hyperplane. These points, and only these points, determine the exact location of H . In this example, there are five support vectors. Using support vectors avoids the curse of dimensionality because all other points are not relevant to the solution.

According to the curse of dimensionality, the more input variables you have, the more data points that you need to fill the space—that is, to determine the true relationship between the inputs and the target.

Question: SVM and the Curse of Dimensionality



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Question: SVM and the Curse of Dimensionality

To construct a support vector machine, only the observations closest to the separating hyperplane are used. This avoids the curse of dimensionality.

- a. True
- b. False

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CA: true

Using only the observations closest to the separating hyperplane avoids the curse of dimensionality by limiting the number of data points in the solution.

Answer: SVM and the Curse of Dimensionality

To construct a support vector machine, only the observations closest to the separating hyperplane are used. This avoids the curse of dimensionality.

- a. True
- b. False

CA: true

Using only the observations closest to the separating hyperplane avoids the curse of dimensionality by limiting the number of data points in the solution.

Essential Discovery Tasks for Support Vector Machines



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Essential Discovery Tasks for Support Vector Machines



Essential Discovery Tasks

- ✓ Select an algorithm.
- Improve the model.
- Optimize the complexity of the model.
- Regularize and tune the hyperparameters of the model.

The essential discovery tasks for support vector machines are the same as for neural networks. Let's build a support vector machine model.

Demo: Building a Support Vector Machine Using the Default Settings

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5.2 Modifying the Model: Methods of Solution



Improving the Support Vector Machine Model



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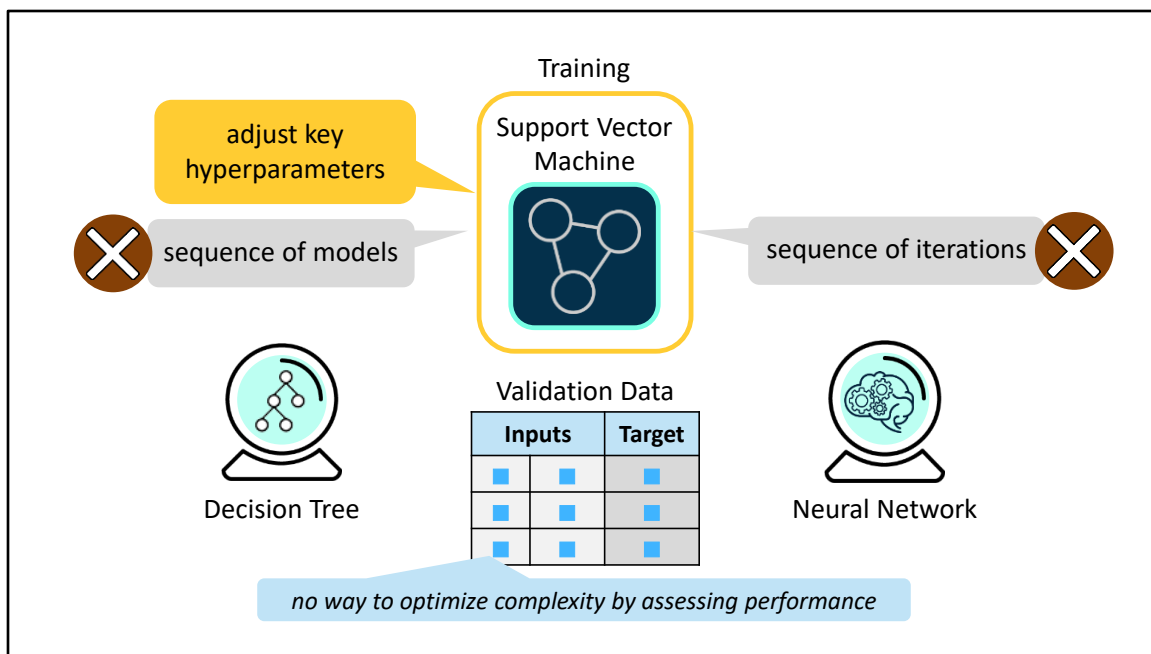
Improving the Support V Machine Model



Essential Discovery Tasks

- ✓ Select an algorithm.
- Improve the model.
- Optimize the complexity of the model.
- Regularize and tune the hyperparameters of the model.

Now that you have built a support vector machine model using the default settings, you are ready to refine it. For support vector machines, there is no clear distinction between the next two discovery tasks: improving the model and optimizing complexity.



Training a support vector machine does not involve a sequence of models, as with decision trees, or a sequence of iterations, as with neural networks. This means that, for support vector machines, there is no way to optimize complexity by assessing performance on validation data across multiple models or iterations.

Instead, you can adjust a few key hyperparameters and then look at the performance of the support vector machine model on validation data.

increase the
model's flexibility

Critical Options in Model Studio

Penalty

a term that accounts for misclassification errors in model optimization

Kernel

a mathematical function that operates as a dot product on transformed data in a higher dimension

Tolerance

a value that balances the number of support vectors and model accuracy

In this lesson, you learn to increase the flexibility of a support vector machine model by modifying the settings of three critical options in Model Studio: penalty, kernel, and tolerance.

The penalty is a term that accounts for misclassification errors in model optimization.

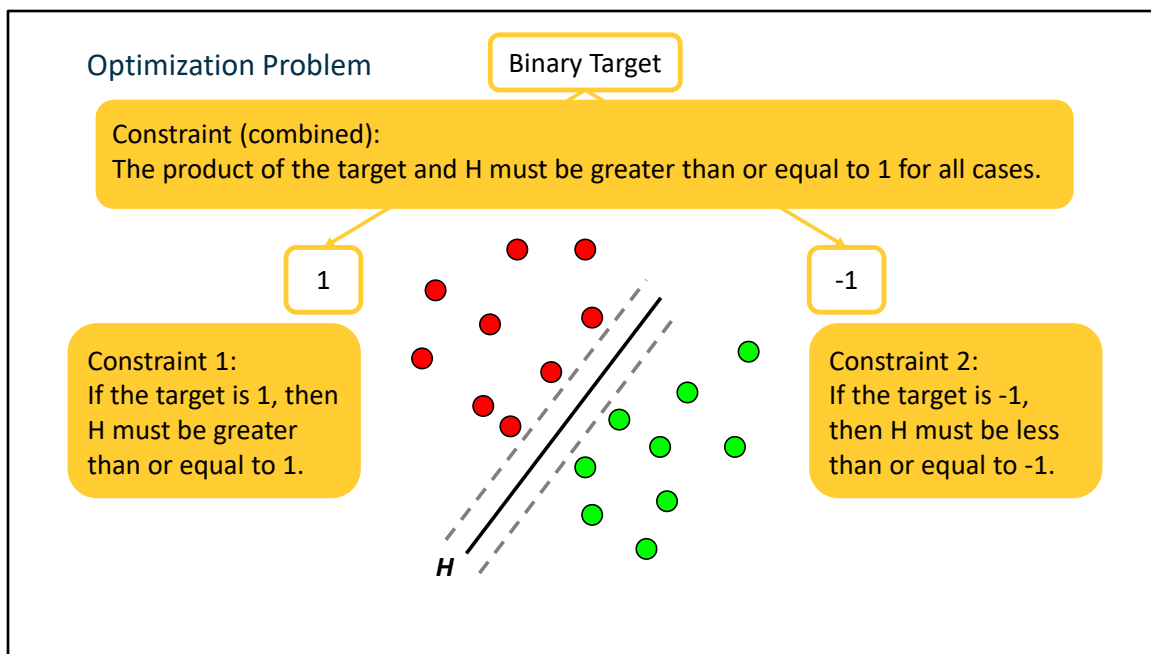
The kernel is a mathematical function that operates as a dot product on transformed data in a higher dimension.

The tolerance value balances the number of support vectors and model accuracy.

Optimization Problem



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To find the best solution to a classification problem, we want to optimize the location of the hyperplane. So, solving for the support vector machine is actually an optimization problem with two constraints.

For mathematical convenience, we define the binary target as either +1 or -1. You learn why a bit later.

The first constraint is that if the target is +1, then the solution for H must be greater than or equal to +1, because it must fall on the correct side of the hyperplane. The second constraint is that if the target is -1, then H must be less than or equal to -1 in order to fall on the correct side of the hyperplane. Because the binary target is defined as +1 or -1, we can combine the two constraints into a single constraint.

The product of the target value and H must be greater than or equal to +1 for all cases. Why? Because we are multiplying either +1 and a value greater than or equal to +1, or -1 and a value less than or equal to -1. In both cases, the product is a positive value.

Constraints for Optimization

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Accounting for Errors with Nonlinearly Separable Data



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Accounting for Errors with Nonlinearly Separable Data

to account for errors use ...

penalty = $C * \text{distance to the hyperplane}$

error weight

soft-margin hyperplane

does make some classification errors

margin size



correct classifications

Trade-Off

small C	larger margin, more misclassified points
large C	smaller margin, fewer misclassified points

H

Our example has now been modified so that the data are no longer linearly separable. When the data are not linearly separable, one solution is a support vector machine known as a soft-margin hyperplane. A soft-margin hyperplane does make some classification errors. It is the analyst's job to decide whether the performance on validation data is acceptable. In a two-dimensional input space, *soft margin* means that a line can separate most of the points but some errors occur.

We account for errors by using a penalty term in the optimization process. This penalty is the product of two quantities: an error weight, which is a regularization parameter often denoted by C , and the distance between a point in error and the hyperplane.

In hard margin SVM, the margin constraints are "hard", and not allowed to be violated at all. You can vary C in order to balance the goals of maximum margin separation, and correctness of the training set classification. You can achieve this trade-off for various values of the parameter C . As C increases we care less about the size of the margin, and more about keeping the error distances small.

Number of Support Vectors

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Lagrange Approach for Estimation

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Demo: Modifying the Methods of Solution Parameters

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Question: Nonlinearly Separable Data



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Question: Nonlinearly Separable Data

When the data are not linearly separable, the process of optimizing the location of the hyperplane must account for classification errors.

- a. True
- b. False

Answer: Nonlinearly Separable Data

When the data are not linearly separable, the process of optimizing the location of the hyperplane must account for classification errors.

- ☒ a. True
- ☐ b. False

CA: a

When the data are not linearly separable, the hyperplane will misclassify some data points. In this situation, the process of optimizing the location of the hyperplane must account for these classification errors.

5.3 Modifying the Model: Kernel Function



Optimizing the Complexity of the Support Vector Machine Model



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Optimizing the Complexity of the Support Vector Machine Model



Essential Discovery Tasks

- ✓ Select an algorithm.
 - Improve the model.
 - Optimize the complexity of the model.
 - Regularize and tune the hyperparameters of the model.
- penalty term
- kernel function

In the previous demonstration, we attempted to refine the model by changing the penalty term. Remember that the penalty term accounts for misclassification errors that inevitably occur when the data are not linearly separable.

Suppose we decide that our model's performance is not yet satisfactory. We want to go beyond a soft-margin classifier.

Next, we apply a kernel function to improve classification.

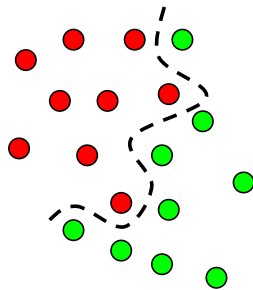
Feature Space Approach for Nonlinearly Separable Data



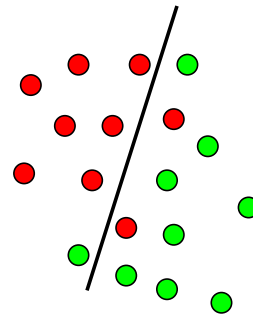
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Feature Space Approach for Nonlinearly Separable Data

Two-Dimensional
Input Space



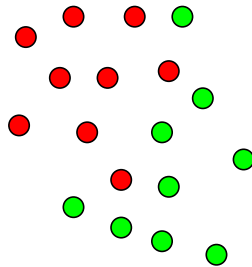
Soft-Margin Classifier



high misclassification rate!

Here is a new, more realistic version of the two-dimensional, red-versus-green example. In this example, even a soft-margin classifier would result in too many errors.

Two-Dimensional Input Space



Feature Space Approach

1. Transform the data to a higher dimension (the feature space).
For example:

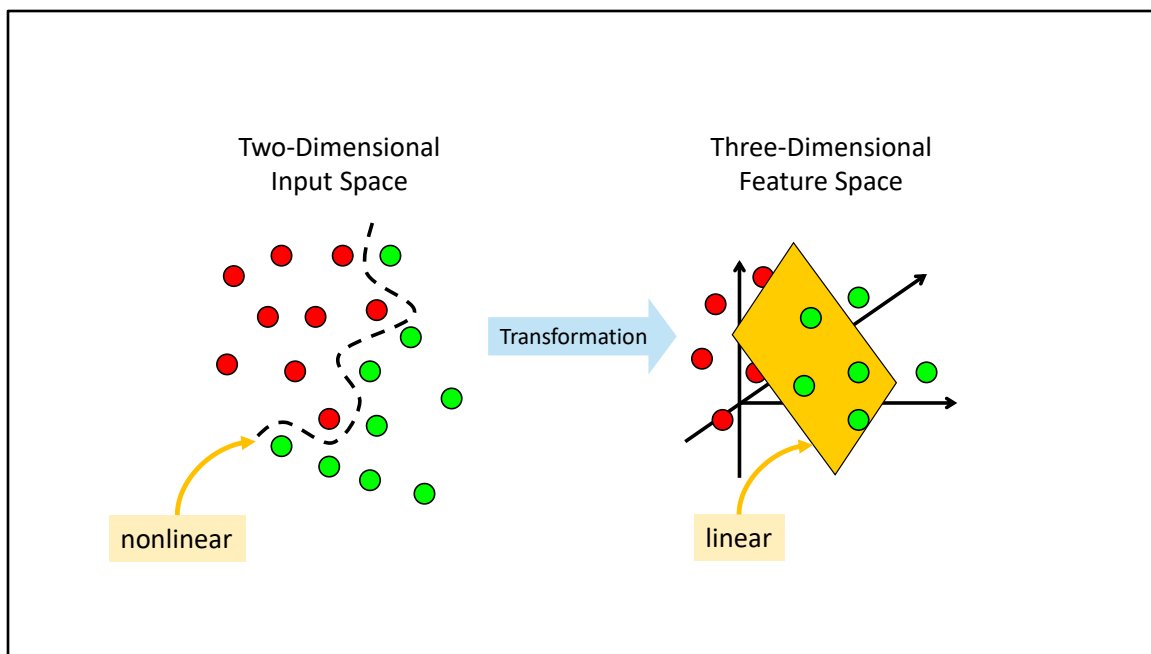
Input space	Feature space
1-D: X	2-D: X, X^2

2. Find the classifier (the maximum-margin hyperplane) in the higher dimension.

Let's look at a different approach to optimizing the complexity of the model: the feature space approach. This approach is viable for most real-world cases.

First, we transform the data – that is, we project the data to a higher dimension using a nonlinear transformation. This higher dimension is called the *feature space*. In a very simple case, with one input, we can transform the input by squaring.

Then we can use two inputs (both the original input and its squared form) to find the classifier (that is, the maximum-margin hyperplane) in this higher dimension.



In the red-versus-green example, we can transform the data into a three-dimensional feature space, where a plane separates the red and green dots.

Kernel Trick



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Kernel Trick

mathematical challenge!

dot product

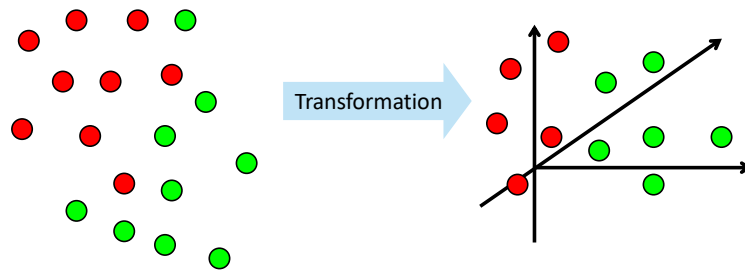
$$\langle x_i, x_j \rangle$$

$$\langle \Phi(x_i), \Phi(x_j) \rangle$$

difficult to calculate

Two-Dimensional
Input Space

Three-Dimensional
Feature Space



Optimizing the complexity of the model in a higher-dimensional feature space presents a mathematical challenge. Dot products are required to solve for H and to score new observations. If the data are transformed to a higher-dimensional feature space, then a dot product is required on the transformed data. A dot product on transformed data has a different form and is difficult to calculate.

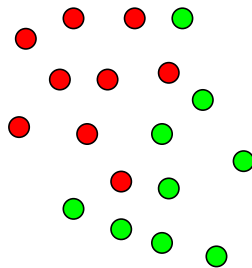
math
trick!

Kernel Function

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$$

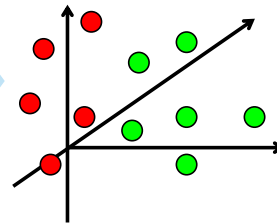
- exists in the original input space
- avoids calculating dot products on transformed data

Two-Dimensional
Input Space



Transformation

Three-Dimensional
Feature Space



To avoid having to calculate dot products on transformed data, we can use a math trick called a kernel function instead. The kernel function exists in the original input space, but it is equivalent to a dot product on the transformed data in the higher-dimensional feature space. We do not need to know the transformation itself! We call this a trick because we do not need to know exactly what the feature space looks like. It is enough to specify the kernel function as a measure of similarity. The geometric interpretation remains the same because the solution is still a hyperplane.

Kernel Function

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$$

Linear

$$k(x_1, x_2) = \langle x_1, x_2 \rangle$$

Polynomial

$$k(x_1, x_2) = (\langle x_1, x_2 \rangle + 1)^p$$

Sigmoid

$$k(x_1, x_2) = \tanh(p_1 \langle x_1, x_2 \rangle + p_2)$$

Radial Basis Function (RBF)

$$k(x_1, x_2) = \exp\left(-\frac{\|x_1 - x_2\|^2}{2\sigma^2}\right)$$

where x_1 and x_2 are two vectors and $\langle, \rangle$ is the inner product

As you'll see in the next demonstration, Model Studio provides two kernel functions, linear and polynomial. Technically, a linear kernel does not apply a transformation to the data. A polynomial kernel raises the data to a power.

Now there are two additional kernels Radial basis function and Sigmoid are available. Sigmoid kernel is equivalent to an input layer activation function used in a multilayer perceptron neural network. RBF is used to perform transformation when there is no prior knowledge about data.

SVM Properties Related to Kernels and Training Algorithms

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Demo: Increasing the Flexibility of the Support Vector Machine

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Question: Kernel Functions



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Question: Kernel Functions

Which of the following statements is true about kernel functions?

- a. A kernel function must be a polynomial.
- b. A kernel function is used to account for errors during the optimization process.
- c. A kernel function operates as a dot product in a higher dimension, but it is applied to the raw data.

Answer: Kernel Functions

Which of the following statements is true about kernel functions?

- a. A kernel function must be a polynomial.
- b. A kernel function is used to account for errors during the optimization process.
- ☒ c. A kernel function operates as a dot product in a higher dimension, but it is applied to the raw data.

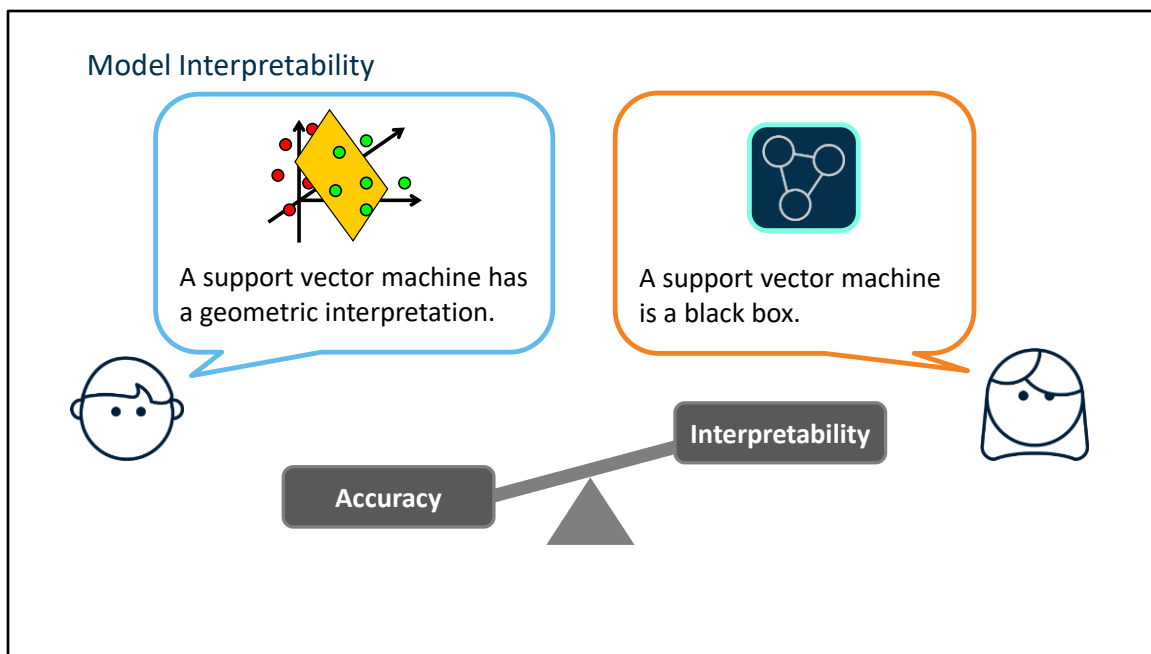
CA: c

A kernel function operates as a dot product in a higher dimension (that is, in a feature space), but it is applied to the raw data.

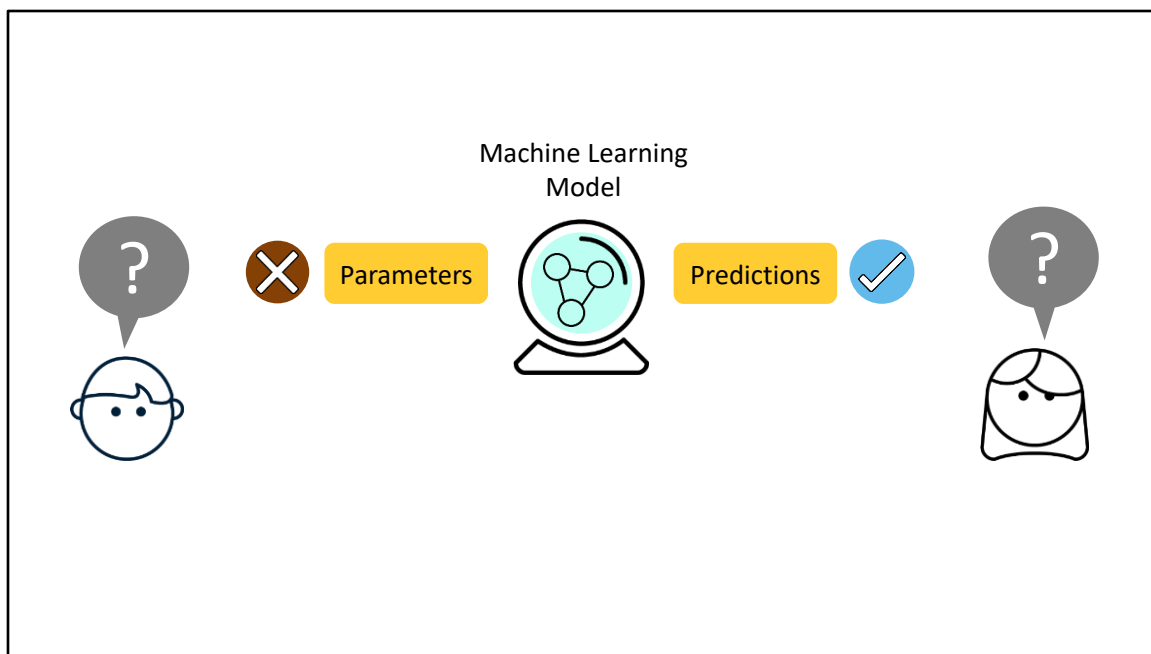
Model Interpretability



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In theory, support vector machines have a geometric interpretation because they are hyperplanes. However, in practice, many analysts consider support vector machine models to be black boxes, much like neural networks. Thus, there is often a trade-off between model accuracy and model interpretability.



Machine learning algorithms are good at generating accurate and generalizable predictive models, but it is often nearly impossible to understand how the algorithms arrived at those predictions or even what the behavior or trend of the model is. We cannot gain much insight into support vector machines, and some other machine learning models, by analyzing the model parameters. However, we can learn a great deal by analyzing the predictions from these models. In the next demonstration, you learn how to use Model Studio's model interpretability feature to interpret your results.

Model Interpretability Plots

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Demo: Adding Model Interpretability

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Question: Model Interpretability



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Question: Model Interpretability

Which of the following machine learning models is the easiest to interpret?

- a. decision tree
- b. neural network
- c. support vector machine

Question: Model Interpretability

Which of the following machine learning models is the easiest to interpret?

- ☒ a. decision tree
- b. neural network
- c. support vector machine

CA: a.

Decision trees are highly interpretable because they are based on English rules, which are rules that use Boolean logic. However, Model Studio's interpretability feature is available for all machine learning models, including decision trees. You might want to use this feature, for example, when you are comparing decision trees to other types of models.

Regularizing and Tuning the Hyperparameters of the Support Vector Machine Model



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Regularizing and Tuning the Hyperparameters of the Support Vector Machine Model



Essential Discovery Tasks

- ✓ Select an algorithm.
- ✓ Improve the model.
- ✓ Optimize the complexity of the model.
- Regularize and tune the hyperparameters of the model.

Autotuning is disabled in SAS Viya for Learners, and selecting it results in an error.

An important discovery task is to regularize and tune the hyperparameters of the model. As with other types of models, we can automate the process of discovering the optimal settings for the support vector machine model. Next, you learn about the hyperparameters for support vector machines that you can autotune in Model Studio.

Autotuning Options for Support Vector Machines in Model Studio

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(Optional) Practice: Build a Support Vector Machine Model Using Autotuning

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5.4 Lesson Conclusion



Lesson Quiz

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